Introduction to Logic and Science
Philosophy 170 Section 3

Fall 2026
Mo & We 10-11:15, Room W506
Office hours: Weds. after class, W1448

Instructor: Boris Ayala
boris.ayala69@cuny.edu

The professor said: "Spicebush swallowtails emerge from their chrysalides only if the temperature doesn't drop below 42 degrees Fahrenheit during pupation. In my lab, I've made sure that throughout the pupation period the temperature never dropped below 42 degrees, so beautiful spicebush swallowtails will soon emerge."

Did you realize that the professor's statement is *wrong*? It's OK if you didn't. The professor made an argument, and many arguments have a peculiar feature: We can evaluate them even when we don't fully understand them. In this course we will learn how to identify and evaluate arguments, even in fields we don't yet master. This will help us craft better arguments and also avoid being misled by bad ones.

Learning Objectives
You will learn how to think well.

Required Text
A Concise Introduction to Logic, Hurley, 14th ed.

Class Philosophy
There is only one way to learn logic, and that is by doing it. To that end, we will have regular homework assignments. These assignments will include questions on material we will review in class *after* the homework is due. So you should make sure to have read, and worked through, the relevant portions of the textbook in advance. You should come to class ready to ask questions. While there will be some lecturing in our course, I expect most of our class sessions to be driven by student questions.

Grade Components

Homework	30 points
Participation	10 points
Test 1	20 points
Test 2	20 points
Final Exam	20 points
Total	100 points

Grade Scale

97 – 100	A+	78 – 80	B-
92 – 96	A	75 – 77	C+
89 – 91	A-	70 – 74	C
85 – 88	B+	60 – 69	D
81 – 84	B	Less than 60	F

Homework
Handwritten homework assignments will be due in class every Wednesday we do not have a test. I will post homework assignments on Brightspace, and each will consist of selected exercises from the textbook. Homework is the biggest single component of your final grade. That's because I want to encourage you to practice logic regularly and also because I aim to reward sustained effort.

You will earn 3 points for every *completed* assignment handed in. On some weeks, I will include bonus exercises for a bonus extra point. All incomplete assignments get 2 points.

I will not accept any late homework. However, only your 10 best assignments will count towards your final grade, so you may miss up to 3 and still get full points. Or more, depending on whether you take advantage of the bonus exercises.

Participation and Attendance
Everybody will begin the semester with 5 points. You can earn up to 5 more by asking questions or volunteering answers. Each day you participate will earn you an extra point. But you may also lose points for things like being on your cell phone or chatting with other students during class. If you don't participate in class, you will keep the original 5 points.

I will take attendance at the beginning of each class. The first three unexcused instances of lateness or absence are free. After that, I will take off one participation point per absence/lateness. However, if you reach six late or absent days in a semester, you will have failed this course. "Late" means *any time after I take attendance*, as I cannot keep track of the exact arrival time of every student.

Tests and Final Exam
The tests and final exam will each last the length of our class session. They are cumulative with an emphasis on more recent material. All tests and the exam will be multiple choice. I will provide the answer forms. *Please make sure you bring at least two pencils and an eraser*. The Scantron machine does NOT read pen marks.

Electronics Policy
The use of any electronics in class is prohibited.

ADA, Sexual Misconduct, and Academic Integrity Policies
This class follows the policies of Hunter College, which you can find on the school's website. I want everyone here to feel safe and welcome. Except for cheaters. Cheating results in failing the course and whatever other consequences Hunter deems fit.

Schedule for the Semester

1	Aug 31 *Welcome*	Sep 2 *Ch. 1.1-1.2, Basic concepts*
2	Sep 7 *College Closed*	Sep 9 *Ch. 1.3-1.4, Deduction and validity*
3	Sep 14 *Ch. 1.5 & Ch. 3.1, Informal fallacies*	Sep 16 *Ch. 4.1-4.2, Categorical prop'ns.*
4	Sep 21 *No Classes*	Sep 23 *Ch. 4.3, Venn diagrams*
5	Sep 28 *Ch. 4.7, Translation*	Sep 30 *Ch. 3.2, Informal fallacies*
6	Oct 5 *Ch. 10, Causality & Mill's methods*	Oct 7 **TEST 1**
7	**TUESDAY** Oct 13 *Ch. 3.3, Informal fallacies*	Oct 14 *Ch. 6.1, Propositional logic*
8	Oct 19 *Ch. 6.1, Symbols and translation*	Oct 21 *Ch. 6.2, Truth functions*
9	Oct 26 *Ch. 6.3, Truth tables*	Oct 28 *Ch. 6.4, More truth tables*
10	Nov 2 *Ch. 3.4, Informal fallacies*	Nov 4 *Ch. 6.6, Argument forms*
11	Nov 9 *Ch. 6.6, More argument forms*	Nov 11 **TEST 2**
12	Nov 16 *Ch. 8.1, Predicate logic*	Nov 18 *Ch. 8.2, Rules of inference*
13	Nov 23 *Ch. 8.3, Quantifier negation*	Nov 25 *No Classes*
14	Nov 30 *Ch. 3.5, Informal fallacies*	Dec 2 *Ch. 11, Probability*
15	Dec 7 *Ch. 12, Statistical reasoning*	Dec 9 *Ch. 13, Hypothetical reasoning*
16	Dec 14 *Ch. 14, Science and superstition*	Dec 16 *Review*
17	Dec 21 – **FINAL EXAM**	